

Impact Statement Card

Team Gambia — TRI AI Saturdays Cohort 10 — Agricultural Extension RAG: Smart Retrieval for Farmers

Intended Positive Impact

An accurate retrieval layer is a prerequisite for any trustworthy agricultural RAG or chatbot system. Improving retrieval quality directly reduces the risk that a farmer receives an answer grounded in the wrong source document — for example, being told to treat a nitrogen deficiency with a potassium-focused intervention, or being pointed to advice for a different crop's pest entirely. In aggregate, this work aims to make downstream agricultural advisory tools more reliable, particularly for smallholder farmers with limited access to in-person extension officers.

Who Benefits

- **Smallholder farmers**, especially those in remote areas with limited access to in-person agricultural extension services, who may rely on mobile-based tools for crop guidance.
- **Agricultural extension officers**, who could use a reliable retrieval tool to speed up triage and response for common queries, freeing time for more complex cases.
- **Developers of downstream RAG systems** in the agricultural domain, who benefit from an openly documented, empirically validated retrieval pipeline and the specific failure patterns (vocabulary mismatch, hard-negative confusion) identified during development.

Potential Risks and Harms

- **Residual retrieval errors causing crop mismanagement:** even a well-tuned system (this pipeline: $nDCG@5 \approx 0.78\text{--}0.87$ depending on evaluation split) does not retrieve the correct document 100% of the time. A wrong retrieval feeding into a generation step could produce confidently wrong agronomic advice, which a farmer may act on without independent verification.
- **Over-reliance on an automated system** in place of professional extension services, particularly for complex or ambiguous cases (e.g. co-occurring stresses) the system was not designed to handle.
- **Uneven performance across crops**, given the corpus's uneven crop distribution (Section: Data Card) — farmers growing under-represented crops (e.g. pearl millet) may receive systematically lower-quality retrieval.
- **Misplaced confidence from evaluation metrics:** a single aggregate $nDCG@5$ score can mask meaningfully worse performance on specific query types or crops; this was observed directly during development (Section: Reproduction / README), where certain query phrasings and crop/soil-topic confusions produced systematic near-zero scores even as the aggregate metric looked strong.

Mitigations Applied in This Project

- Retrieval-only scope: this project deliberately does not build or claim to build a full advisory chatbot; it isolates and evaluates the retrieval component so its limitations are visible and measurable rather than hidden inside an end-to-end system.
- Per-query error analysis (not just aggregate metrics) was used throughout development to surface specific, systematic failure patterns rather than relying on a single summary number.
- Cross-validation (5-fold) was used to report an honest performance range rather than a single potentially-overfit number, and known gaps between local and leaderboard evaluation are documented transparently in the project README rather than omitted.

Recommended Safeguards for Any Deployment

- Any production system built on this retrieval component should include a clear disclaimer that outputs are not a substitute for professional agronomic advice, and should provide an easy path to a human extension officer for cases involving uncertainty or high stakes (e.g. suspected disease outbreaks with economic significance).
- Retrieval confidence should be surfaced to any downstream generation step, so low-confidence retrievals can trigger a fallback (e.g. "I'm not confident about this — here are the closest matches, please verify with a local extension officer") rather than a confidently-generated but ungrounded answer.